The Milky Way Through the Lens of CPTG

Galactic Rotation, Material Structure, and Geometric Gravity

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Abstract

The Milky Way is our home galaxy, a barred spiral system whose stellar disk, gas layers, central bulge and extended outskirts can be investigated from within. This vantage point makes its gravitational structure unusually accessible, although the motion of the Sun and the geometry of the disk must be accounted for when reconstructing Galactic rotation. We examine the Milky Way through Curvature Polarization Transport Gravity (CPTG), which retains Einstein gravity and ordinary stellar and gas matter while describing an additional gravitational response through coupled curvature polarization and transport. The study proceeds from 45 published Gaia DR3+ composite circular-speed measurements across 5.25–27.25 kpc to an independently specified finite-height McMillan material model, a corrected and numerically verified oblate-bulge force, and the registered CPTG reconstruction of the Galactic circular field. The result has $\chi^2_{\mathrm{diag}} = 7.347858$ and a velocity root-mean-square residual of $1.719\,\mathrm{km\,s^{-1}}$. At 27.25 kpc, the reconstructed gravitational mass equivalent is $2.079304 \times 10^{11}\,M_\odot$, comprising $6.499360 \times 10^{10}\,M_\odot$ of independently integrated physical material and $1.429368 \times 10^{11}\,M_\odot$ of geometric mass equivalent, a geometric share of **68.7426**%. Jiao et al.'s independent Gaia DR3 Galactic B2 model assigns **69.9512**% of its fitted dynamical mass to an inferred dark halo, a difference of 1.2086 percentage points between the reported shares. At the common radius of 27.25 kpc, their published Einasto profile encloses $1.436575 \times 10^{11}\,M_\odot$, versus CPTG's $1.429368 \times 10^{11}\,M_\odot$ geometric equivalent. A downstream measurement of the completed CPTG field gives the Milky Way Structural Mode $N_{\mathrm{MW}} = 2.826813$, with an organization length of 9.639830 kpc. Together these measurements connect the observed rotation, the distribution of physical matter, the reconstructed geometric gravitational response and the radial organization of our Galaxy.

1 Introduction

The Milky Way is the galaxy in which the Solar System resides. It is a large, rotating, barred spiral system, with a flattened stellar disk crossed by spiral structure, layers of atomic and molecular gas, a thicker stellar population extending above and below the main disk, and a concentrated central bulge. These components are not distributed in a sphere. They occupy different regions of the Galaxy, and their gravitational influence depends both on how much matter they contain and on where that matter lies. The Galactic disk also exhibits warping and other departures from ideal axial symmetry, especially at large radii [11, 4]. An account of Milky Way gravity must therefore pay attention to the geometry of its source as well as to the motions of its stars.

We observe this galaxy from a particularly unusual position: the Sun moves inside its disk rather than viewing it from outside. Nearby stellar distances, directions and motions can be measured in great detail, allowing the material and dynamical structure to be investigated together. Yet the same internal vantage point makes a rotation curve harder to obtain than a simple photograph of a distant, inclined galaxy might suggest. Each measurement is made relative to the moving Sun; dust obscures parts of the disk; and a star's measured velocity includes both its Galactic orbit and motions that depart from ideal circular rotation. A Galactic rotation curve is therefore a reconstructed dynamical measurement, assembled using distances, proper motions, line-of-sight speeds and the adopted Galactic reference frame.

The quantity of interest is the *circular speed* $V(R)$, the speed that would provide circular support at distance R from the Galactic center in the inferred gravitational field. Circular motion connects the measured speed to an inward acceleration through $g(R) = V^2(R)/R$. Thus a rotation curve tells us how much radial gravitational acceleration is required as we move outward through the disk. A second, independent question asks how much of that acceleration follows from the observed populations of stars and gas. Because most Galactic matter lies in a flattened distribution, calculating its gravitational force requires more than placing an enclosed mass at the center of a sphere.

Curvature Polarization Transport Gravity (CPTG) supplies the framework used here to connect these questions [6]. CPTG retains the Einstein description of gravity and the stress–energy of ordinary matter, and includes coupled geometric polarization and transport responses in the gravitational field. In its registered galaxy calculation, the measured rotation profile and the constructed material source are used to recover the complete circular field. Once that field is obtained, one can separately describe its gravitational mass equivalent, compare that quantity with the integrated physical matter, and measure how the resulting field is organized across radius. Structural Mode N is the latter measurement: an object-specific fingerprint taken from the completed field, not an input to its calculation [9].

The prior CPTG investigation, *CPTG in 153 Primary SPARC Galaxies: Resolved Rotation Curves and Geometric–Material Mass Equivalents*, reported individual rotation curves and material–geometric mass equivalents for 153 primary SPARC galaxies at 3,168 measured radii [8]. It established the resolved six-curve presentation used here. The Milky Way provides a distinctive companion case: its material source can be reconstructed as an

explicit finite-height Galactic density distribution, while its rotation is inferred from observations made inside the moving disk.

This study uses the published 45-point DR3+ Galactic rotation compilation of Sylos Labini et al. [1], which combines results from Eilers et al. [2] and Wang et al. [3]. The material source is taken from the finite-height stellar and gas distributions of McMillan (2017) [4], with the central bulge treated in its oblate geometry. A separate Gaia DR3 mass analysis by Jiao et al. [11] supplies a numerical comparison with a conventional inferred halo profile.

The paper first explains how the Galactic rotation measurements are obtained and how the stars and gas are represented. We then introduce the CPTG field equation and its specific galaxy reduction, describe the numerical reconstruction, and read the resulting rotation curve and mass profiles. Finally, we measure Structural Mode N and compare the CPTG geometric mass equivalent with the Gaia-based halo analysis. Throughout, the source matter, the total gravitational field and the post-solution diagnostics are kept individually identifiable.

2 Observed Rotation and Galactic Material Structure

2.1 Published DR3+ circular-speed input

A radius in the present analysis is a *Galactocentric* distance, measured outward from the Galactic center in the plane of the modeled disk, rather than a distance measured from Earth. The observations first locate and track stars relative to the Solar System. Gaia astrometry supplies positions, parallaxes and proper motions; spectroscopy or related surveys supply velocities along the line of sight. After transformation to the Galactic frame, the kinematics of stellar populations are analyzed to infer the circular speed required by the local gravitational field. The latter is a dynamical inference from the stellar motions, not simply the measured speed of one particular star.

We adopt the 45 DR3+ Galactocentric radius and circular-speed pairs in Table 1 of Sylos Labini et al. [1], together with their quoted speed uncertainties. For $i = 0, \ldots, 44$, the sampled radii are

$$R_i = (5.25 + 0.5i)\,\mathrm{kpc}. \tag{1}$$

The first reported point is $226.8 \pm 1.9\,\mathrm{km\,s^{-1}}$ and the last is $176.9 \pm 8.2\,\mathrm{km\,s^{-1}}$. Each uncertainty indicates the quoted uncertainty in the inferred circular speed at that radius. The measured range therefore follows a substantial portion of the stellar disk and its outer regions, rather than describing only the Solar neighborhood.

The DR3+ sample combines two published dynamical determinations based on Gaia-linked observations. Their reductions account for the solar and Galactic reference frame and use dynamical assumptions, including approximate axisymmetry and treatments of noncircular stellar motions and velocity dispersion [2, 3, 1]. These are part of the measurement procedure that produces the published rotation curve. We retain its radii,

circular speeds and uncertainties without re-estimating the stellar kinematics. Later, the quoted uncertainties will determine how the residuals between the observed and reconstructed speeds are summarized.

2.2 Stellar disks, interstellar gas, and bulge

A rotation curve specifies the gravitational acceleration needed for circular support; it does not itself reveal how the Galaxy's physical matter is distributed. To calculate that distribution separately, we use the material components of McMillan's Galactic density model [4]. The *thin stellar disk* describes the relatively concentrated layer of stars around the Galactic plane, while the *thick disk* represents a more vertically extended stellar component. Neutral atomic hydrogen (H I) and molecular gas form additional, differently distributed layers. The central bulge is compact and flattened. The source is thus a collection of spatial densities rather than a single spherical mass.

The source model provides both a force and a mass inventory, which are different outputs. Integrating its density over a chosen spatial region counts the physical material there. Solving for the gravitational field of that density determines the acceleration at a chosen location. For a flattened disk, matter outside a given cylindrical radius can also influence the local radial force, so the force cannot in general be recovered from the material mass interior to that radius using a spherical shortcut.

The vertical extent of each disk is retained in the midplane radial-force calculation. For a representative exponential stellar disk, the density is

$$\rho_d(R,z) = \frac{\Sigma_{0,d}}{2h_d} \exp\left(-\frac{R}{R_d} - \frac{|z|}{h_d}\right), \tag{2}$$

Here $\rho_d(R,z)$ is the local mass density, $\Sigma_{0,d}$ sets the disk's surface-density normalization, R_d controls its radial falloff, and h_d specifies the height over which material is distributed around the plane $z = 0$. The gas components use their own profiles, including the prescribed central suppression. Table 1 lists the adopted scales and normalizations. These material parameters are fixed inputs from the Galactic model; the rotation-curve reconstruction below does not use them as adjustable Galactic masses.

Component	Σ_0 ($M_\odot\,\mathrm{kpc}^{-2}$)	R_d (kpc)	h_d (kpc)
Thin stellar disk	8.95679×10^8	2.49955	0.300
Thick stellar disk	1.83444×10^8	3.02134	0.900
H I gas disk	5.31319×10^7	7.000	0.085†
Molecular gas disk	2.17995×10^9	1.500	0.045†
Bulge: oblate spheroid, $q = 0.5$, $r_0 = 0.075\,\mathrm{kpc}$, cutoff $r_{\mathrm{cut}} = 2.1\,\mathrm{kpc}$			

Table 1: Adopted finite-height Galactic material source. Surface-density and scale parameters are those of the archived McMillan-model configuration. The gas profiles also carry their prescribed inner-cutoff parameters. †*Positive physical scale-height magnitudes; the source configuration encodes the gas profile using a negative-height convention.*

The bulge illustrates why its geometry matters. A spherical bulge would treat all directions about the Galactic center equally, whereas an *oblate* bulge is flattened along its short axis. Its density is organized by $s^2 = R^2 + z^2/q^2$, with flattening $q = 0.5$. The inward radial acceleration in the Galactic plane must therefore be computed using the oblate homeoidal integral

$$g_{R,\mathrm{bulge}}(R,0) = \frac{4\pi G q}{R} \int_0^R \frac{\rho_{\mathrm{bulge}}(s)\, s^2\, ds}{\sqrt{R^2 - (1-q^2)s^2}}. \tag{3}$$

The denominator preserves the radial geometry of flattened density shells. To verify this step independently, the same specified material distribution was also evaluated with a two-dimensional axisymmetric weak-form Poisson solver, which computes the gravitational field from the density on a numerical grid. Across all 45 sampled radii, the greatest relative difference between the two calculations of the total midplane material acceleration is approximately 0.0036% on the finest verification grid. This agreement establishes the numerical consistency of the adopted source-force calculation before it enters CPTG.

Figure 1 lets the reader see what each material component contributes. A curve is plotted in circular-speed units by converting its radial acceleration to the speed corresponding to that acceleration at radius R. The individual curves are visualizations of forces from stars, gas and bulge; their signed accelerations, rather than their displayed speeds, form the total material force.

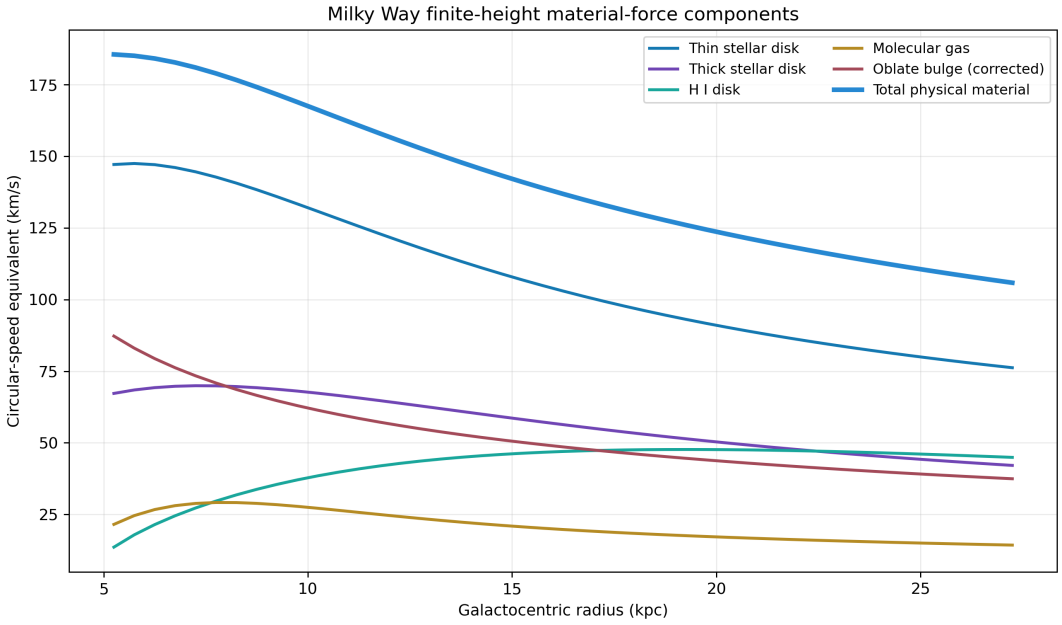


Figure 1: What the modeled Galactic matter contributes to circular support. Each curve translates the independently calculated midplane radial force of a stellar, gas or bulge component into the corresponding circular-speed equivalent. The blue total-material line is formed from the signed sum of the component accelerations. The source is three-dimensional, with finite disk heights and a corrected oblate bulge.

3 The CPTG Gravitational Framework

3.1 Coupled field and registered radial reduction

In General Relativity, the curvature of spacetime and the stress–energy of matter are linked by Einstein's field equation. CPTG begins with that same gravitational structure and retains ordinary matter as its physical source. It adds two coupled geometric contributions: *curvature polarization*, describing the organized geometric response, and *curvature transport*, describing the corresponding transport response across the gravitational configuration [6]. In the theory, these terms act through the geometry rather than by adding a new material density to the stellar and gas source.

The parent CPTG field equation expresses that arrangement as

$$G_{\mu\nu} + \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} = \frac{8\pi G}{c^4} T^{\mathrm{material}}_{\mu\nu}. \tag{4}$$

The subscript pairs μ, ν label components of spacetime tensors. The Einstein tensor $G_{\mu\nu}$ remains present throughout the calculation; $T^{\mathrm{material}}_{\mu\nu}$ describes ordinary stellar and gas matter. The tensors $\Pi^{\mathrm{pol}}_{\mu\nu}$ and $\Theta^{\mathrm{trans}}_{\mu\nu}$ denote the coupled geometric polarization and transport terms. When those terms vanish, the same field equation reduces to Einstein's equation. This makes General Relativity a continuously retained part of the CPTG description.

For galaxies, the registered numerical procedure works with the radial acceleration implied by circular motion, rather than directly solving the entire four-dimensional tensor equation for the Galaxy. That practical reduction introduces several quantities with different jobs. The result we want is $g_C(R) = V_C^2(R)/R$, the total reconstructed circular acceleration. The quantity $g_b(R)$ is the signed, unit-normalized source template supplied to the registered galaxy solver. Finally, $a_\star$ is a local acceleration scale calculated from the measured rotation profile, used to set the observation-conditioned structural support at each sampled radius. In the notation employed for earlier SPARC galaxy reconstructions [7, 8, 5],

$$g_b(R) = \frac{Q(V_{\mathrm{gas}}) + Q(V_{\mathrm{disk}}) + Q(V_{\mathrm{bulge}})}{R}, \qquad Q(V) = V|V|, \tag{5}$$

$$a_\star(R_i) = \frac{V^2_{\mathrm{obs},i}}{R_i}. \tag{6}$$

The operation $Q(V) = V|V|$ keeps the sign of each input force contribution while expressing it in the velocity-equivalent notation used by the galaxy workbench. That sign matters when a modeled component's radial contribution changes direction. The unit-template source g_b is the source passed to the registered CPTG equation. The finite-height acceleration g_{mat}, calculated from McMillan's physical densities, is a separate source-field readout. The main figure displays both so that the role of the solver's source normalization remains visible to the reader.

CPTG's radial reduction also responds to *how the field changes with radius*, not only to its value at a given point. The observed support profile is smoothed to obtain $g_{\mathrm{sup}}(R)$

and the radial derivatives needed by the registered structural and polarization terms. The dimensionless Σ_C^2 describes the local support shape, $\mathcal{C}_C$ is the carrier constructed from the field and its radial derivatives, and B_{pol} represents the resulting polarization term. With the registered conventions [7],

$$\Sigma_C^2(R) = \frac{1}{3}\left[1 - \frac{d\ln g_{\mathrm{sup}}}{d\ln R}\right] \geq 0, \tag{7}$$

$$\mathcal{C}_C(R) = g_{\mathrm{sup}}^2 + \kappa_C^2\left[(g'_{\mathrm{sup}})^2 + 2(g_{\mathrm{sup}}/R)^2\right], \tag{8}$$

$$B_{\mathrm{pol}}(R) = \frac{(R^2 + 2\kappa_C^2) g_{\mathrm{sup}}\,\mathcal{C}_C^{-2/3} - \frac{d}{dR}\left[R^2\kappa_C^2 g'_{\mathrm{sup}}\mathcal{C}_C^{-2/3}\right]}{R^2}. \tag{9}$$

The local support shape is kept at its calculated nonnegative value, including values above unity. The polarization term then enters the *implicit* radial field equation below. Implicit means the unknown total field g_C appears on both sides; its physical numerical value must be found by solving the equation, rather than obtained from one direct substitution:

$$\left(1 + c_C\Sigma_C^2\right) g_C = g_b + A_C a_\star^{2/3} B_{\mathrm{pol}}\left[1 + \left(\frac{g_C}{a_\star}\right)^{123/50}\right]^{-11/50}. \tag{10}$$

The term on the left organizes the completed circular field together with the support-shape contribution. On the right, the signed material template and the polarization expression are coupled through a prescribed suppression factor. The fractional powers and all source conventions are retained from the registered galaxy law. At each sampled radius, $a_\star$ is computed from the published circular speed, and the implicit equation is solved for the completed field. Three coefficients control the registered radial reconstruction for the Galaxy and are optimized against the same published speed measurements. Their fitted values are

$$\kappa_C = 22.534704\,\mathrm{kpc}, \qquad A_C = 6.962575 \times 10^{-6}, \qquad c_C = 111.406414, \tag{11}$$

Here κ_C sets a radial scale within the carrier and polarization operator, while A_C and c_C enter the strength of the registered response terms; A_C is stated in the workbench's numerical convention. The published speeds enter both the local support $a_\star$ and the fit for these coefficients. The resulting rotation curve is therefore an *observation-conditioned reconstruction*: it measures how the registered CPTG law organizes the observed Galactic field and its specified source. The fitted coefficients describe the combined response, with sensitivity to the adopted observational support and its radial smoothing.

4 Methodology: Reconstructing the Milky Way

The computational sequence follows the physical reasoning just developed. Each operation has a specific input and output, so that the observed speeds, the material source, the reconstructed field and the later diagnostics can be checked separately.

4.1 Preparing the two observational and material inputs

First, the 45 published radius–speed pairs and their quoted errors are retained on the original half-kiloparsec grid. They provide the observational circular field and, through $V_{\rm obs}^2/R$, the local support acceleration needed by the registered galaxy method. The independently specified material-density input enters in the following step.

Second, the three-dimensional McMillan density model specifies the physical stellar disks, gas layers and oblate bulge. The sampled midplane forces of the four disk and gas components enter the main reconstruction as signed, precomputed force contributions. The oblate-bulge force is evaluated directly from Equation (3). Their sum defines the physical material-force profile; a separate integration of the adopted densities yields the material inventory. The independent two-dimensional axisymmetric Poisson calculation evaluates the density source on a grid and checks the assembled midplane field across all 45 radii.

4.2 Physical-force inputs and registered source templates

The physical material-force curve and the baryonic source template in the CPTG reconstruction are constructed from the same set of modeled Galactic components but use distinct numerical normalizations. Let $g_{R,j}(R,0)$ denote the signed inward radial acceleration of component j in the midplane, and define $u_j(R) = R\,g_{R,j}(R,0)$ in $(\mathrm{km\,s^{-1}})^2$, with R expressed in kpc. The sampled values $u_{\rm thin}$, $u_{\rm thick}$, $u_{\rm HI}$, and $u_{\rm H_2}$ are supplied in the evidence file `inputs/milky_way_45.csv` as the four `v...2_km2_s2` columns. These are precomputed midplane force contributions for the adopted finite-height McMillan source, not four freely fitted mass coefficients. The bulge acceleration $g_{R,\rm bulge}$ is calculated at each sampled radius from Equation (3) and the specified oblate density.

The registered source-template convention applies fixed divisors to the material-force equivalents: unity for gas, 0.5 for the combined stellar disks, and 0.7 for the bulge. Using the signed-square operation $Q(V) = V|V|$ introduced in Equation (5), the input transformations are

$$\begin{aligned} Q(V_{\rm gas}) &= u_{\rm HI} + u_{\rm H_2}, \\ Q(V_{\rm disk}) &= \frac{u_{\rm thin} + u_{\rm thick}}{0.5}, \\ Q(V_{\rm bulge}) &= \frac{R\,g_{R,\rm bulge}}{0.7}. \end{aligned} \tag{12}$$

The physical radial acceleration, in contrast, uses the modeled force contributions without these source-template divisors:

$$g_{\mathrm{mat}}(R) = \frac{u_{\mathrm{thin}} + u_{\mathrm{thick}} + u_{\mathrm{HI}} + u_{\mathrm{H_2}}}{R} + g_{R,\mathrm{bulge}}(R, 0). \tag{13}$$

Table 2 identifies the numerical fields used in this transformation. The gas velocity template preserves the sign of its summed force, while the sampled stellar-disk and bulge terms are inward. The registered 0.5, 1, and 0.7 factors are fixed source-template conventions carried over from the galaxy solver; they are not new Galactic parameters selected by the 45-point fit. For the CPTG field equation, the acceleration g_b formed from Equation (5) is converted to SI units, using $1\,\mathrm{kpc} = 3.085677581491367 \times 10^{19}\,\mathrm{m}$.

Source component	**Sampled force data**	**Registered input**
Thin and thick stellar disks	`vthin2_km2_s2`, `vthick2_km2_s2`	`vdisk_unit_km_s`: $Q(V_{\mathrm{disk}}) = (u_{\mathrm{thin}} + u_{\mathrm{thick}})/0.5$
Atomic and molecular gas	`vHI2_km2_s2`, `vH2_2_km2_s2`	`vgas_signed_km_s`: $Q(V_{\mathrm{gas}}) = u_{\mathrm{HI}} + u_{\mathrm{H_2}}$
Oblate bulge	Density parameters in `inputs/McMillan17.ini`; Equation (3)	$Q(V_{\mathrm{bulge}}) = R\, g_{R,\mathrm{bulge}}/0.7$

Table 2: Physical-force sampling and fixed baryonic template inputs used by the CPTG numerical reconstruction. All u_j values and signed-square velocity contributions are in $(\mathrm{km\,s^{-1}})^2$. The actual input is assembled from the registered disk and gas columns together with the calculated oblate-bulge field.

The finite-height disk and gas force samples may also be checked directly from the source densities by solving the axisymmetric Poisson equation, $\nabla^2\Phi = 4\pi G\rho$, and sampling the positive-inward convention $g_R = \partial\Phi/\partial R$ in the midplane (the outward-coordinate acceleration is $-\partial\Phi/\partial R$). The supplied `axisymmetric_weak_field.py` performs this independent density-to-force calculation. The principal `run.py` calculation reads the tabulated disk and gas force samples, evaluates the bulge integral and integrates the density to obtain the material aperture masses. The independent force solver reproduces the combined modeled midplane acceleration to approximately 0.0036% over the 45 radii. The sampled source fields and the density parameters are both included in the numerical evidence package, allowing either branch of the material-force check to be inspected.

4.3 Solving for the circular gravitational field

Third, the material component templates are assembled with their registered signed-source convention. The uncertainty-weighted observational support curve and its radial derivatives supply $a_\star$, Σ_C^2 and the polarization carrier. At every radius the nonlinear equation is solved for the admissible g_C; the three galaxy-level coefficients are selected by the workbench's registered optimization across the full 45-point sample. The result is one completed circular-field profile with a corresponding rotation speed $V_C(R) = \sqrt{R\, g_C(R)}$.

Fourth, the calculated speeds are placed beside the published measurements. Differences in speed are reported directly and through a diagonal-error χ^2 statistic. This is an in-sample evaluation of the registered reconstruction because the published velocities participate

in the support and coefficient calculation. This explicit dependence identifies the result as an observation-conditioned reconstruction and documents how the circular field was obtained.

4.4 Reading the completed result

Fifth, the completed circular field is converted to a spherical gravitational mass equivalent. The independently integrated material inventory is subtracted to obtain the geometric mass-equivalent profile, preserving its distinction from the local finite-height material force. The same completed field is then passed to the separate Structural Mode operator to measure its radial organization. Lastly, the CPTG mass equivalent is compared at specified radii with Jiao et al.'s Gaia-based fitted halo mass. These final operations read out the solved field; they do not alter it.

4.5 Physical inventory and gravitational equivalents

To understand the mass comparison, it helps to distinguish *mass counted from density* from *mass inferred from circular gravity*. The first is the material inventory $M_{\rm material}(< R)$ obtained by integrating the physical density of stars and gas within the adopted aperture. The second asks what mass would give the same circular acceleration if represented by a spherical equivalent, $M^{\rm eq}_{\rm grav}(< R) = RV_C^2/G$. CPTG's geometric mass equivalent is the difference between these two constructions. The radial accounting is

$$M^{\rm eq}_{\rm grav}(< R) = \frac{RV_C^2(R)}{G}, \tag{14}$$

$$M^{\rm eq}_{\rm geom}(< R) = M^{\rm eq}_{\rm grav}(< R) - M_{\rm material}(< R), \tag{15}$$

$$f^{\rm eq}_{\rm geom}(R) = \frac{M^{\rm eq}_{\rm geom}(< R)}{M^{\rm eq}_{\rm grav}(< R)}. \tag{16}$$

The material inventory retains the disk-and-bulge geometry used in the source integration. The spherical gravitational equivalent is a standard way to express circular acceleration in mass units. Its residual after subtracting the separately counted matter is named the *geometric mass equivalent* in CPTG. At each radius the identity $M^{\rm eq}_{\rm grav} = M_{\rm material} + M^{\rm eq}_{\rm geom}$ makes the accounting transparent. The geometric equivalent is reported in mass units because those units allow a direct comparison with conventional dynamical descriptions of the Galactic field.

5 The Milky Way Rotation Curve

A useful first question for any gravitational reconstruction is straightforward: how closely does the calculated circular speed follow the measured one at each radius? Figure 2 answers that question visually, and Equation (17) provides a numerical summary. For

each of the 45 radii, the difference between the CPTG speed and the published speed is divided by its quoted uncertainty; the squares of those normalized differences are added. The resulting diagonal-error statistic is

$$\chi^2 = \sum_{i=1}^{45} \left(\frac{V_C(R_i) - V_{\mathrm{obs},i}}{\sigma_i} \right)^2 = 7.347858, \tag{17}$$

The velocity root-mean-square difference is $1.719\,\mathrm{km\,s^{-1}}$, providing a summary in familiar speed units. The acceleration comparison is also reported logarithmically as $\Delta_g = \log_{10}(g_C/g_{\mathrm{obs}}) = 2\log_{10}(V_C/V_{\mathrm{obs}})$. Across the 45 bins its population standard deviation is 0.007519 dex and its RMS is 0.007980 dex. These figures describe the scatter of the reconstructed acceleration around the acceleration inferred from the same measured rotation curve.

The central visual result is Figure 2, which follows the visual conventions of the 153-primary-galaxy CPTG atlas [8]. The blue points and error bars are the observed circular speeds, and the orange line is the completed CPTG rotation curve. The black line shows the unit-template baryonic source that enters the registered radial equation. The blue solid line is the *physical* finite-height material-force curve computed from McMillan's densities. The red line converts the geometric mass equivalent into circular-speed units for visual comparison. The green line is the fixed-a_0 illustrative MOND prescription evaluated with the registered source template. These curves answer different questions: measured speed, reconstructed speed, solver input, independently modeled material force and mass-equivalent response. Their velocities are not added as separate parts of the orange curve.

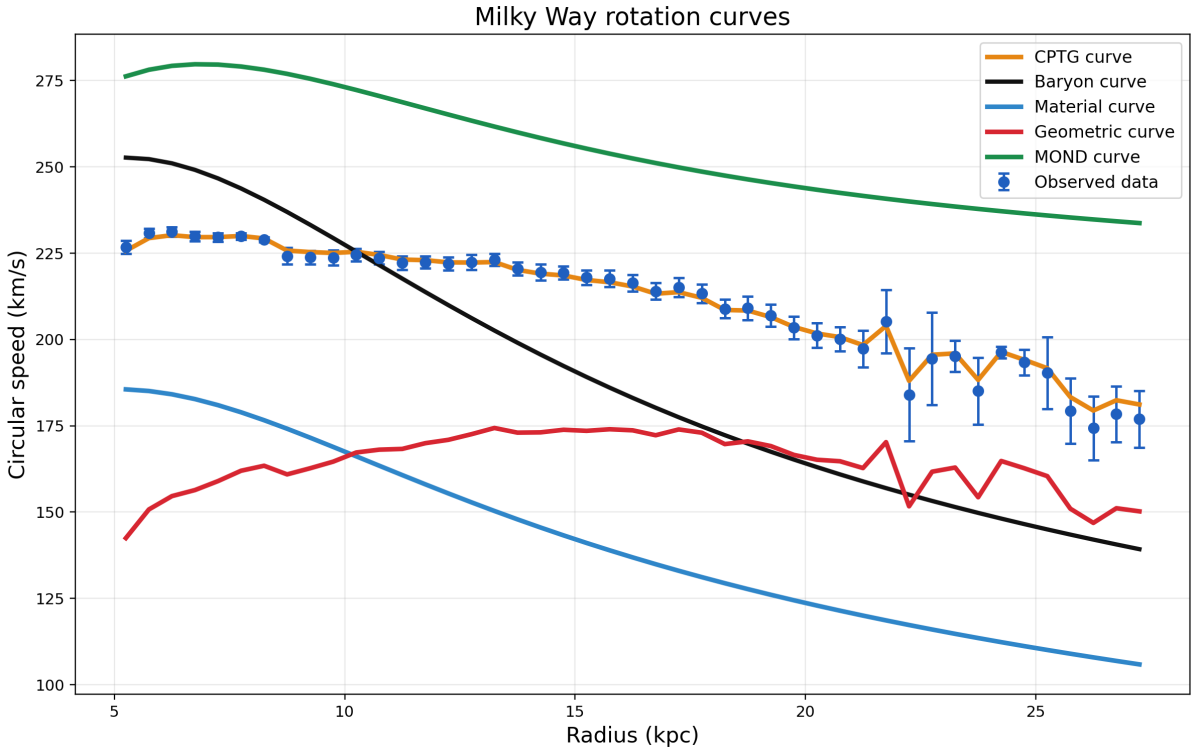


Figure 2: Six views of the Milky Way rotation measurement. Blue points with error bars: published DR3+ circular speeds. Orange: CPTG reconstruction of the circular field. Black: registered unit-template baryonic source supplied to the CPTG solver. Blue solid line: independently calculated finite-height physical material-force speed equivalent. Red: velocity corresponding to the geometric mass equivalent, $\sqrt{GM^{\rm eq}_{\rm geom}(<R)/R}$. Green: fixed-$a_0 = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ illustrative MOND curve using the registered source template. The colored curves represent different measurements and constructions; the physical-force and mass-equivalent velocities are not linearly additive.

The measured rotation decreases across the outer part of the sampled Galactic disk, and the reconstructed curve follows that decline. At 27.25 kpc, the completed CPTG speed is $181.1575\,\mathrm{km\,s^{-1}}$ beside the published $176.9 \pm 8.2\,\mathrm{km\,s^{-1}}$. The lines connect the sampled calculations rather than introducing additional observational bins. Table 3 lets the reader follow selected radii from the observed speed to the reconstructed speed and then to the physical material and geometric mass equivalents.

R	$V_{\rm obs}$	V_C	$V_{\rm mat}$	$M_{\rm material}$	$M^{\rm eq}_{\rm geom}$	$f^{\rm eq}_{\rm geom}$
5.25	226.80	225.70	185.54	3.7411	2.4770	39.84
8.25	228.90	229.27	176.61	4.9609	5.1222	50.80
12.25	222.00	222.35	155.39	5.7602	8.3208	59.09
16.25	216.40	215.36	136.86	6.1263	11.3966	65.04
20.25	201.20	201.75	122.90	6.3221	12.8413	67.01
27.25	176.90	181.16	105.85	6.4994	14.2937	68.74

Table 3: Corrected Milky Way reconstruction at selected radii. Radii are in kpc, all speeds in $\mathrm{km\,s^{-1}}$, masses in $10^{10} M_\odot$, and the geometric-equivalent fraction in percent. $V_{\rm mat}$ denotes the physical finite-height material-force speed equivalent.

6 Material and Geometric Mass Equivalents

The rotation curve has now supplied an acceleration profile, and the material model has supplied a physical density inventory. Expressing both in mass units makes their radial relation visible. Figure 3 follows three quantities outward through the Milky Way: the spherical equivalent of the completed CPTG gravity, the amount of ordinary matter integrated from the spatial density model, and the remaining geometric equivalent. Across the sampled outer disk, the modeled material inventory grows more slowly than the field-equivalent mass. At the last measured radius, $R = 27.25\,\mathrm{kpc}$, the resulting accounting is

$$M_{\mathrm{grav}}^{\mathrm{eq}} = 2.079304 \times 10^{11}\, M_{\odot}, \tag{18}$$

$$M_{\mathrm{material}} = 6.499360 \times 10^{10}\, M_{\odot}, \tag{19}$$

$$M_{\mathrm{geom}}^{\mathrm{eq}} = 1.429368 \times 10^{11}\, M_{\odot}, \tag{20}$$

$$f_{\mathrm{material}} = 31.2574\%, \qquad f_{\mathrm{geom}}^{\mathrm{eq}} = 68.7426\%. \tag{21}$$

At 27.25 kpc, the physical material inventory supplies 31.2574% of the spherical equivalent of the completed field, and the geometric residual supplies 68.7426%. These percentages are defined for the sampled Galactic aperture and its reconstructed circular acceleration. Their explicit radius matters: the composition of the equivalent changes as one moves through the Galactic disk, as the selected-radii table and mass-profile figure demonstrate.

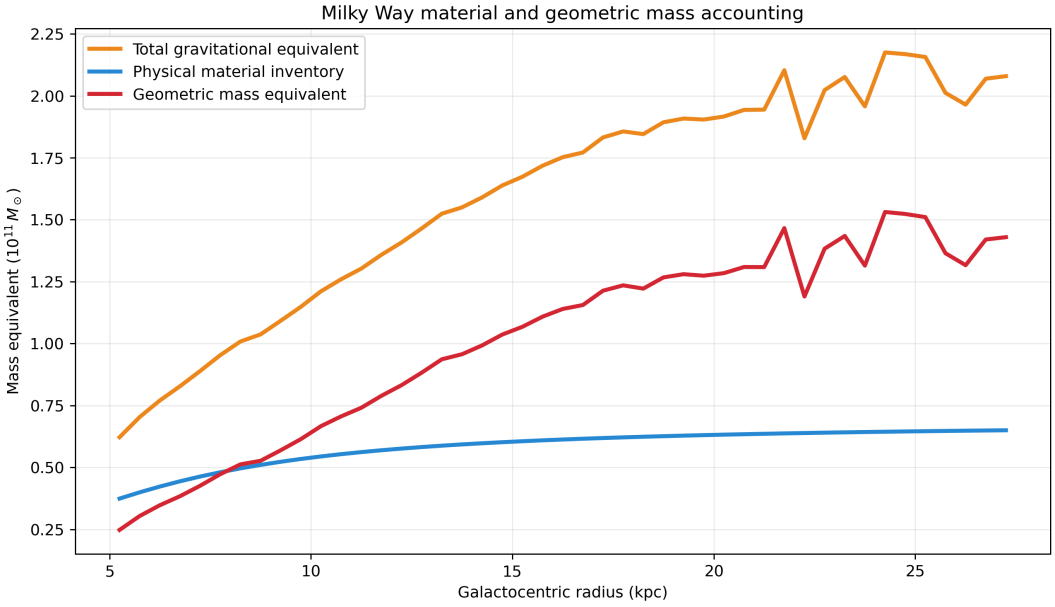


Figure 3: Radial gravitational mass-equivalent accounting for the corrected 45-point CPTG reconstruction: total circular-field spherical equivalent RV_C^2/G (orange), independently integrated finite-height physical material inventory (blue), and their defined geometric mass-equivalent difference (red). The total gravitational equivalent follows a spherical circular-field convention, while the material inventory uses the axisymmetric Galactic source.

This is also why the two material quantities in the figures deserve separate labels. The local midplane force of a flattened disk is calculated from the entire axisymmetric density

distribution; the material inventory simply counts matter inside an aperture. The red geometric curve represents a spherical equivalent formed after the CPTG circular field is known. It is the mass quantities, not the plotted velocities, that satisfy the accounting identity in Equation (15).

6.1 How the mass-equivalent shares respond to the input measurements

A mass-equivalent percentage contains information from two separately determined quantities: the completed radial gravitational field and the physical material inventory. At a specified radius, the former is proportional to the square of the reconstructed circular speed, while the latter is an integral of the adopted stellar and gas densities. This allows the sensitivity of the percentage to each input to be displayed directly. Holding the radius fixed, let δ_v be a fractional change in the reconstructed speed and δ_m a fractional change in the integrated material inventory. The resulting geometric share is

$$f^{\mathrm{eq}}_{\mathrm{geom}}(\delta_v, \delta_m) = 1 - \frac{M_{\mathrm{material}}(1+\delta_m)}{M^{\mathrm{eq}}_{\mathrm{grav}}(1+\delta_v)^2}. \tag{22}$$

This relation is exact for the specified changes to those two quantities; it introduces no new gravitational coefficient. Table 4 evaluates independent illustrative variations about the corrected 27.25 kpc baseline. Each row holds the other quantity at its recorded value, allowing the reader to see separately the response to the recovered gravitational speed and to the physical material inventory.

Specified input change at 27.25 kpc	**Geometric share (%)**
Corrected registered reconstruction	68.7426
Material inventory increased by 5%	67.1797
Material inventory decreased by 5%	70.3055
Reconstructed circular speed increased by 1%	69.3585
Reconstructed circular speed decreased by 1%	68.1080

Table 4: Direct sensitivity of the outer CPTG mass-equivalent share to separately prescribed changes in reconstructed circular speed and adopted physical material inventory. The 1% and 5% cases are numerical sensitivity steps, not assigned measurement errors or new model fits.

The published rotation-point errors enter the registered galaxy reconstruction through its observational support and coefficient fitting. In particular, the outermost published speed is $176.9 \pm 8.2\,\mathrm{km\,s^{-1}}$. A statistical uncertainty in the completed CPTG share requires those correlated observational inputs to be carried through the full reconstruction, together with the uncertainties of the independently adopted Galactic material density model. The present calculation records the central reconstruction and the explicit sensitivities in Table 4, keeping them distinguishable from a joint covariance estimate.

7 Structural Mode N of the Milky Way

Two galaxies can have similar circular speeds yet distribute the changes in their gravitational fields over very different radial regions. A comparison based only on gravitational

strength does not describe that organization. In CPTG, *Structural Mode N* is a dimensionless measurement of where the completed gravitational field changes most strongly across the available radial extent [9]. It is measured *after* the circular field has been reconstructed, so it does not select the mass source, alter the fitted coefficients or determine the rotation curve.

For the Milky Way we start with the completed $g_C(r) = V_C^2(r)/r$ from the preceding sections. The registered structural carrier combines the squared field with a term depending on its radial slope. Taking the magnitude of the carrier's radial change then gives a nonnegative structural weight. The galaxy operator is

$$C_N(r) = g_C^2(r) + \kappa_{\mathrm{struct}}^2 \left(\frac{dg_C}{dr} \right)^2, \qquad \kappa_{\mathrm{struct}} = 1\,\mathrm{kpc}, \tag{23}$$

$$w_N(r) = \left| \frac{dC_N}{dr} \right|, \qquad N_{\mathrm{gal}} = \frac{R}{\lambda_N}. \tag{24}$$

The weight w_N emphasizes regions where the carrier changes appreciably. Its weighted average radius is the *organization length* λ_N: the characteristic radial location of the organized change in this measurement. Structural Mode compares the available outer radius R with that length through $N = R/\lambda_N$. Here R is 27.25 kpc, the outer edge of the sampled Galactic field. The corresponding continuous centroid would be

$$\lambda_{N,\mathrm{cont}} = \frac{\displaystyle\int_{R/5}^{R} r\, w_N(r)\, dr}{\displaystyle\int_{R/5}^{R} w_N(r)\, dr}. \tag{25}$$

The registered sampled galaxy operator and its radial structural interval are used without modification [9]. The actual calculation uses the registered *sampled* version of this operator, evaluated at the published radial bins rather than inferred from an unmeasured continuous profile. Radial differences first determine $D_r g$ at each point (one-sided at the endpoints and centered inside the range). The sampled carrier is $C_i = g_i^2 + \kappa_{\mathrm{struct}}^2 (D_r g)_i^2$, and its change supplies $u_i = |(D_r C)_i|$. We retain the 44 radii beyond $R/5 = 5.45$ kpc and apply the prescribed five-point average $\mathcal{S}_5$, with zero extension at the ends of that retained array. The resulting weighted-radius calculation is

$$w_i = (\mathcal{S}_5 u)_i, \qquad \lambda_{N,\mathrm{num}} = \frac{\sum_i r_i w_i}{\sum_i w_i}, \qquad N_{\mathrm{MW}} = \frac{R}{\lambda_{N,\mathrm{num}}}. \tag{26}$$

The registered sums use the sampled weights as written, without extra bin-width factors. Applying that same operator to the completed Milky Way field gives

$$N_{\mathrm{MW}} = 2.826813, \qquad \lambda_{N,\mathrm{num}} = 9.639830\,\mathrm{kpc}, \qquad R = 27.25\,\mathrm{kpc}. \tag{27}$$

The largest sampled structural weight occurs at 6.75 kpc, while the weighted centroid lies farther out at 9.639830 kpc because the structural profile extends across the observed interval. Figure 4 displays this distinction between a local peak and an overall characteristic radius. The resulting $N_{\mathrm{MW}} = 2.826813$ is the Milky Way's individual Structural Mode

fingerprint over the present sampled range 5.25–27.25 kpc. It is obtained from the Galactic field alone, without using a galaxy classification as a numerical input.

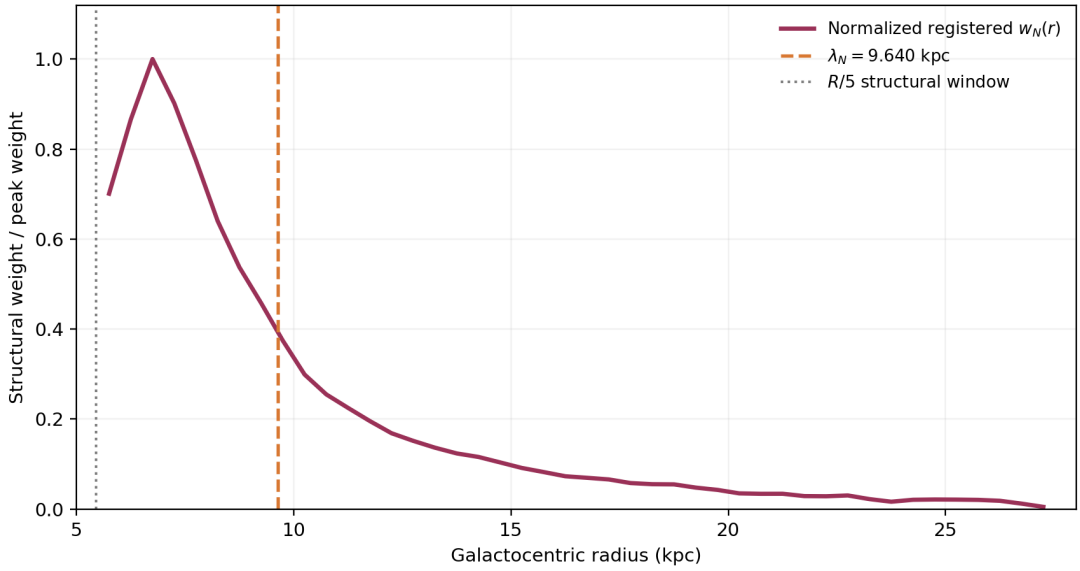


Figure 4: Milky Way Structural Mode measured downstream from the corrected, completed CPTG radial acceleration. The curve is the registered five-point-smoothed outer structural weight w_N normalized to its maximum; the dashed line is the sampled centroid $\lambda_N = 9.639830$ kpc, and the dotted line marks the $R/5 = 5.45$ kpc outer-window boundary. The resulting fingerprint is $N_{\mathrm{MW}} = 2.826813$ for the sampled $R = 27.25$ kpc container.

7.1 Sampling and derivative sensitivity of Structural Mode

The registered N readout uses a finite set of radii and numerical derivatives of the completed acceleration field. Its sensitivity can therefore be measured by repeating only this *downstream* operation, leaving the underlying CPTG velocities and fitted coefficients unchanged. Table 5 shows the result for the registered prescription, two changes in derivative-weight processing, and two shorter measurement containers evaluated from the same corrected field. The outer-window condition $r > R/5$ is reapplied whenever the container radius changes.

Post-solution structural calculation	N
Registered five-point weight, $R = 27.25$ kpc	2.826813
Unsmoothed weight, $R = 27.25$ kpc	2.926689
Second-order edge derivatives, $R = 27.25$ kpc	2.824511
Registered weight, shorter container $R = 26.25$ kpc	2.730165
Registered weight, shorter container $R = 25.25$ kpc	2.790581

Table 5: Sampling and operator sensitivity of the Milky Way post-solution Structural Mode. The registered five-point result remains the reported Galactic fingerprint; other rows show the effect of explicitly stated numerical alternatives and radial coverage, without any CPTG field refitting.

These calculations identify two practical aspects of measuring Galactic organization. The derivative convention changes the recorded five-point result only slightly at the full radius,

while smoothing and the choice of outer observational container have larger effects. A future independently measured extension of the Milky Way rotation curve will supply a new outer container for the same prescribed structural operator.

8 Comparison with the Gaia-Inferred Dark Halo

The Milky Way's declining rotation curve has also been analyzed within the conventional description of ordinary matter plus an inferred dark halo. Jiao et al. [11] used Gaia DR3 stellar kinematics, explored several baryonic models and fitted a spherical Einasto halo. In that approach, the halo represents the portion of the gravitational mass attributed to the dark component after accounting for the assumed stars and gas. Comparing its quantitative mass budget with CPTG is especially informative because both investigations address the outer Milky Way using the published Galactic rotation measurements, while reporting their gravitational components in different frameworks.

Jiao et al.'s abstract gives a representative dynamical mass of $2.06^{+0.24}_{-0.13} \times 10^{11}\, M_\odot$. Their Table 4 specifies a B2 material model with $M_{\rm dyn} = 2.05^{+0.08}_{-0.06} \times 10^{11}\, M_\odot$ and $M_{\rm bar} = 0.616 \times 10^{11}\, M_\odot$. Subtracting those printed central values gives the mass assigned to the inferred halo:

$$M^{\rm B2}_{\rm halo,Jiao} \simeq M^{\rm B2}_{\rm dyn} - M^{\rm B2}_{\rm bar} = 1.434 \times 10^{11}\, M_\odot. \tag{28}$$

Dividing the inferred halo mass by the B2 dynamical total expresses that result as a percentage, just as the CPTG calculation expressed its geometric equivalent as a percentage of its total gravitational equivalent:

$$f^{\rm B2}_{\rm DM,Jiao} = \frac{M^{\rm B2}_{\rm dyn} - M^{\rm B2}_{\rm bar}}{M^{\rm B2}_{\rm dyn}} = \frac{1.434}{2.050} = 69.9512\%. \tag{29}$$

The B2 dynamical total in Jiao et al. is reported with asymmetric bounds of $+0.08$ and -0.06 in units of $10^{11}\, M_\odot$. Keeping the printed B2 baryonic mass fixed while inserting the dynamical bounds into Equation (29) gives a dark-matter share of 69.0452% for $1.99 \times 10^{11}\, M_\odot$ and 71.0798% for $2.13 \times 10^{11}\, M_\odot$. This is the B2 dynamical-mass contribution to the share's uncertainty, calculated from the published bounds and conditional on the listed baryonic model.

CPTG's corresponding geometric share is 68.7426% at 27.25 kpc (Equation (21)). The two percentages differ by 1.2086 percentage points. The Jiao figure describes the B2 model's fitted Galactic mass partition, using its virial-halo convention; the CPTG figure describes the reconstructed mass-equivalent partition at the last measured radius. Naming these mass conventions makes both published results intelligible in the same table.

A further comparison can be made at an identical physical radius, rather than using only each study's headline mass totals. Jiao et al.'s inferred halo is represented by an *Einasto density profile*, a smooth three-parameter function describing how their spherical halo density falls with distance from the center. Integrating that density inside a sphere of radius r gives its enclosed halo mass. We evaluate Jiao et al.'s published B2 central

parameters, with $\rho_{\rm Ein}(r) = \rho_0 \exp[-(r/h)^{1/n}]$ and normalization $M_0 = 4\pi h^3 \rho_0$, to obtain

$$M_{\rm halo,Jiao}(< r) = M_0 n\, \gamma\!\left(3n, \left(\frac{r}{h}\right)^{1/n}\right), \quad (M_0, h, n) = (3.72 \times 10^{11}\, M_\odot, 11.41\,{\rm kpc}, 0.43), \tag{30}$$

Here $\gamma(a, x)$ denotes the lower incomplete gamma function arising from integration of the Einasto profile, and (M_0, h, n) are the rounded central parameters printed by Jiao et al. At 27.25 kpc, that profile contains $1.436575 \times 10^{11}\, M_\odot$ in inferred halo mass. CPTG gives $1.429368 \times 10^{11}\, M_\odot$ as its geometric mass equivalent at the same radius, a difference of 0.50% relative to Jiao's enclosed profile.

The published rounded quantities permit a second arithmetic check. Jiao et al.'s B2 tabulated dynamical-minus-baryonic mass is $1.434000 \times 10^{11}\, M_\odot$. Integrating the separately rounded B2 Einasto parameters to the study's reported representative virial radius of 121.03 kpc gives $1.438107 \times 10^{11}\, M_\odot$, a difference of $0.004107 \times 10^{11}\, M_\odot$, or 0.286% of the tabulated residual. The profile mass at 27.25 kpc exceeds the printed tabulated residual by $0.002575 \times 10^{11}\, M_\odot$ (0.180%). The table totals and profile parameters were published at different numerical precision; the radial comparison consistently uses the explicitly evaluated Einasto profile, and the percentage comparison consistently uses the printed B2 mass totals.

Jiao et al.'s Table 4 also reports individual parameter ranges for the fitted Einasto normalization, scale and index. Varying one published B2 parameter across its printed asymmetric range while keeping the other two at their central values produces the enclosed masses shown in Table 6. This makes the profile's parameter sensitivity visible without imposing an unreported joint parameter covariance.

Varied B2 parameter	**Published endpoints**	$M_{\rm halo}(< 27.25\,{\rm kpc})$ $(10^{11} M_\odot)$
M_0 $(10^{11} M_\odot)$	3.02, 4.17	1.1663, 1.6104
h (kpc)	9.79, 12.56	1.4380, 1.4315
n	0.34, 0.55	1.2507, 1.7921

Table 6: One-parameter sensitivity of the Jiao B2 enclosed Einasto halo at the common radius 27.25 kpc. The input ranges are Jiao et al.'s separately published asymmetric parameter endpoints; in each row the two unlisted parameters retain their central B2 values. The paired masses follow the endpoint order shown in the middle column.

Mass definition and source	Mass	Share (%)
Jiao et al. B2 total dynamical mass	2.050	100.0000
Jiao et al. B2 physical baryonic mass	0.616	30.0488
Jiao et al. B2 inferred dark matter	**1.434**	**69.9512**
Jiao B2 Einasto halo inside 27.25 kpc	1.436575	—
CPTG total gravitational equivalent at 27.25 kpc	2.079304	100.0000
CPTG physical material inventory at 27.25 kpc	0.649936	31.2574
CPTG geometric mass equivalent at 27.25 kpc	**1.429368**	**68.7426**

Table 7: Gaia-based B2 dark-matter mass share and CPTG geometric mass-equivalent share. Jiao et al.'s B2 percentages use their published fitted Galactic dynamical and baryonic totals; CPTG percentages use the reconstructed mass equivalents at 27.25 kpc. The separate Einasto entry is the Jiao B2 halo integrated to that same radius. Masses are in $10^{11}\,M_\odot$; B2 percentages are arithmetic values derived from the rounded masses printed in Jiao et al. Table 4.

The radial curves convey more information than one endpoint percentage. At 19.25 kpc, CPTG's geometric equivalent is $1.279974 \times 10^{11}\,M_\odot$ versus $1.354496 \times 10^{11}\,M_\odot$ for the B2 Einasto halo, a 5.50% lower CPTG value. At 26.25 kpc the values are 1.316244 and 1.435292 in the same units, an 8.29% lower CPTG value. The curves cross elsewhere in the plotted interval. Figure 5 therefore provides a radius-by-radius view of where the two recovered gravitational components are most alike and where their radial structure differs.

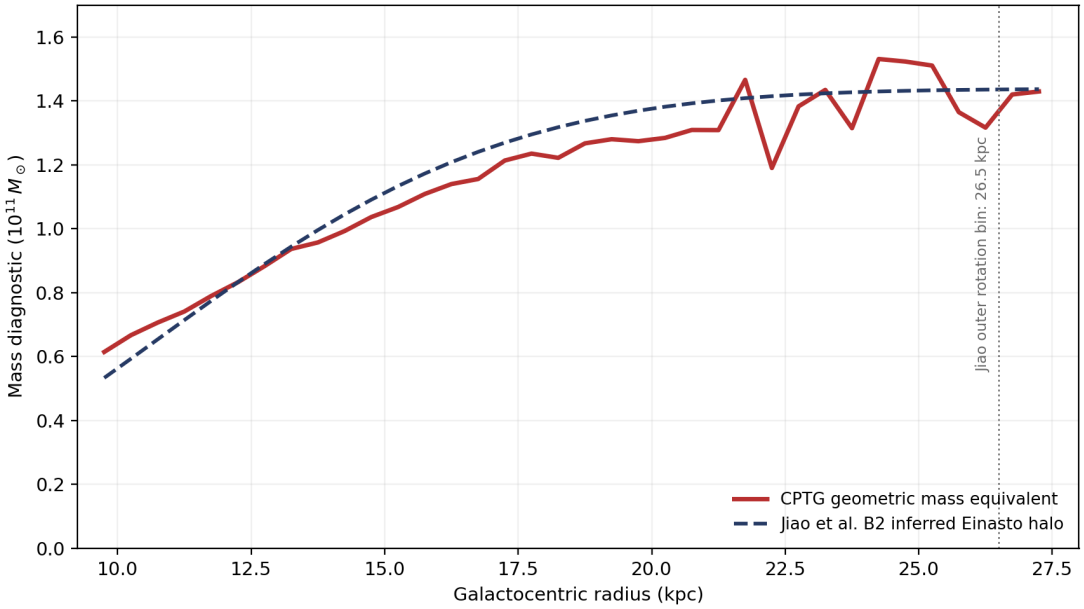


Figure 5: Radial comparison between CPTG's geometric mass-equivalent residual (red, reconstructed circular-field equivalent minus independently integrated physical material inventory) and the spherical enclosed Einasto halo calculated from Jiao et al.'s published B2 central parameters (navy dashed). The vertical marker locates Jiao et al.'s last observed circular-speed bin at 26.5 kpc; their fitted halo profile is evaluated through 27.25 kpc to display both mass constructions at the same outer radius.

These comparisons bring the reader back to the central physical distinction in the paper. The CPTG calculation uses the DR3+ composite velocities and a McMillan finite-height density model to reconstruct the circular field and its geometric mass equivalent. Jiao et

al. use their Gaia DR3 dynamical analysis, B2 baryonic source and Einasto density model to infer a halo mass. Both mass budgets are calculated from explicitly stated inputs; the table compares the reported shares, and the radial figure evaluates the published halo profile and CPTG equivalent at common radii.

9 Discussion

The Milky Way offers an unusually rich setting for CPTG because several measurements that are often compressed into one external-galaxy rotation curve can be kept separate. Stellar kinematics determine a sampled circular gravitational field; observed stellar and gas distributions inform a three-dimensional material model; and that material model can be checked independently with an axisymmetric gravitational solver. The completed CPTG field is then read in several ways: as a rotation curve, as radial gravitational mass equivalents, and as an individual Structural Mode fingerprint. The purpose of reporting all of them is to make the gravitational account of our own Galaxy as physically transparent as possible.

One central result is the corrected finite-height source force. The stellar disks, gas layers and flattened bulge are evaluated according to their geometry. The independent axisymmetric Poisson calculation verifies their combined midplane acceleration across the 45 radii to approximately 0.0036% on the finest grid. This source-field check is carried out before the CPTG fitting step and anchors the physical material side of the reconstruction.

The registered CPTG field follows the declining outer-Galaxy rotation profile, with a diagonal-error statistic $\chi^2 = 7.347858$ and RMS speed residual of $1.719\,\mathrm{km\,s^{-1}}$. At 27.25 kpc, the same field corresponds to $2.079304 \times 10^{11}\,M_\odot$ in spherical mass equivalent. The independent physical inventory is $6.499360 \times 10^{10}\,M_\odot$; the geometric remainder is $1.429368 \times 10^{11}\,M_\odot$, or 68.7426% of that equivalent. These quantities show directly how the recovered gravitational response varies alongside the Galaxy's modeled physical material.

Structural Mode adds a different dimension to this description. It does not measure how massive the Milky Way is or how fast one representative star travels. By following the change of the completed field over radius, it locates the characteristic organization length $\lambda_N = 9.639830\,\mathrm{kpc}$ and returns the sampled Galactic fingerprint $N_{\mathrm{MW}} = 2.826813$. The reported mode is specific to the present 27.25-kpc radial container and the completed reconstruction used to measure it.

The Gaia-based comparison supplies an independent way to quantify the outer gravitational account. Jiao et al.'s B2 analysis assigns 69.9512% of its fitted Galactic dynamical mass to inferred dark matter; the CPTG mass accounting gives 68.7426% geometric equivalent at its measured outer boundary. In mass units, integrating Jiao et al.'s published Einasto model to 27.25 kpc gives $1.436575 \times 10^{11}\,M_\odot$, beside CPTG's $1.429368 \times 10^{11}\,M_\odot$. The 0.50% endpoint difference is accompanied by the explicitly plotted radial variations.

9.1 Reading the observational and model assumptions

Several inputs explain how this result was obtained and indicate what additional observations will be most informative. The DR3+ speeds come from a composite of published kinematic analyses, using distance estimates, solar-reference-frame choices and an axisymmetric treatment of stellar motions [1, 2, 3]. Their quoted errors supply the diagonal weights used in the fit statistic. Measurements of vertical motions and other noncircular kinematics offer complementary information about the Galaxy beyond this radial circular-speed profile.

The McMillan material distribution likewise specifies stellar and gas density scales drawn from a broader Galactic modeling program [4]. Its disk thicknesses, bulge flattening and material normalization determine both the independent force profile and the mass inventory. Equation (22) quantifies the response of the 27.25 kpc geometric share to separately varied field and material inputs. The detailed McMillan material-parameter covariance and the covariance of the reconstructed Gaia composite are distinct inputs to a future joint uncertainty calculation.

The post-solution Structural Mode sensitivity in Table 5 records the effects of sampled coverage, smoothing and derivative conventions. Likewise, the published Jiao B2 dynamical-mass bounds and Einasto parameter intervals yield the explicit conditional mass-share and radial-profile variations reported in the comparison section. In each case, the central result and the stated variation are identified by their input assumptions.

The registered CPTG reconstruction uses the observed velocities to define local $a_\star$, the smoothed support profile and three fitted coefficients. Consequently, the stated χ^2 measures consistency of this *observation-conditioned* reconstruction with its input rotation data. The derivative support and coefficient trade-offs have been examined in separate numerical stress tests. Those results identify useful targets for future checks with independent kinematic bins, vertical acceleration constraints and refined observations of the outer material distribution.

The present calculation connects a physically specified three-dimensional source to a radial gravitational reconstruction. Completing independent off-plane gravitational measurements and extending well-characterized kinematic data beyond the current radius would add new constraints on the full Galactic field and on the radial reach of its Structural Mode measurement.

10 Conclusion

The Milky Way is a Galactic laboratory in which the motions of stars and the spatial distribution of the matter sourcing gravity can be studied together. In this work, 45 published DR3+ circular-speed measurements were paired with an independently specified finite-height material model containing thin and thick stellar disks, atomic and molecular gas, and an oblate bulge. The corrected physical source force agrees with an independent two-dimensional axisymmetric Poisson calculation to approximately 0.0036% at the sampled radii.

The registered CPTG reduction reconstructs the observed Galactic field with

$$\chi^2_{\mathrm{diag}} = 7.347858, \qquad \mathrm{RMS}(V_C - V_{\mathrm{obs}}) = 1.719\,\mathrm{km\,s^{-1}}. \tag{31}$$

At 27.25 kpc, its spherical gravitational mass equivalent is $2.079304 \times 10^{11}\,M_\odot$. The physical material inventory is $6.499360 \times 10^{10}\,M_\odot$, leaving $1.429368 \times 10^{11}\,M_\odot$ in geometric mass equivalent, or **68.7426**% of the completed field equivalent at that radius.

The completed field also supplies a Milky Way Structural Mode $N_{\mathrm{MW}} = 2.826813$, with an organization length $\lambda_N = 9.639830\,\mathrm{kpc}$. In the independent Gaia DR3 analysis of Jiao et al., the B2 model assigns **69.9512**% of its fitted dynamical mass to inferred dark matter, a difference of 1.2086 percentage points from CPTG's outer geometric share. Evaluated at the common radius of 27.25 kpc, the published B2 Einasto profile gives $1.436575 \times 10^{11}\,M_\odot$ in inferred halo mass beside CPTG's $1.429368 \times 10^{11}\,M_\odot$ geometric equivalent.

Together the rotation curve, verified material force, mass-equivalent profiles and Structural Mode describe the Milky Way in terms of both the strength and radial organization of its gravitational field. The same framework allows the effects of improved Galactic kinematics and physical material measurements to be followed explicitly through each stage of the reconstruction.

Data, Code, and Evidence Availability

The complete numerical evidence supporting this paper is supplied in the public CPTG repository [10]:

`CPTG_Milky_Way_Evidence_Package.zip`

The package contains the 45 Gaia DR3+ circular-speed measurements, McMillan Galactic material-density parameters, CPTG gravitational reconstruction code, and executable scripts for reproducing and verifying the results. It includes the axisymmetric material-force calculation, integrated physical material inventory, reconstructed rotation curve, radial gravitational and geometric mass equivalents, and Structural Mode N measurement.

The package also contains the Jiao et al. Gaia-based mass comparison, Einasto radial calculations, numerical sensitivity analyses, and data used to generate the paper's figures. Reproducibility instructions, calculation logs, numerical output tables, verification results, and SHA-256 checksums are provided to independently reproduce and inspect the reported findings.

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